

Problem 1

Solution:

(a) For the base case, observe that $1 = \frac{1(2)}{2}$. For the inductive hypothesis, assume that for arbitrary fixed integer $k \geq 1$, $1 + 2 + \cdots + k = \frac{k(k+1)}{2}$. Now, for the inductive step,

$$\begin{aligned} 1 + 2 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \quad (\text{by inductive hypothesis}) \\ &= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2} \end{aligned}$$

(b) For the base case, observe that $1 = 2(1) - 1$. For the inductive hypothesis, assume that for arbitrary fixed integer $k \geq 1$, $(2(1) - 1) + (2(2) - 1) + \cdots + (2k - 1) = k^2$. Now, for the inductive step,

$$\begin{aligned} 1 + 3 + \cdots + (2k - 1) + (2(k+1) - 1) &= k^2 + (2(k+1) - 1) \quad (\text{by inductive hypothesis}) \\ &= k^2 + 2k + 1 \\ &= (k+1)^2 \end{aligned}$$

(c) For the base case, observe that $1^3 = 1 = 1^2$. For the inductive hypothesis, assume that for arbitrary fixed integer $k \geq 1$, $1^3 + 2^3 + \cdots + k^3 = (1 + 2 + \cdots + k)^2$. Now, for the inductive step,

$$\begin{aligned} 1^3 + 2^3 + \cdots + k^3 + (k+1)^3 &= (1 + 2 + \cdots + k)^2 + (k+1)^3 && (\text{by inductive hypothesis}) \\ &= \left(\frac{k(k+1)}{2} \right)^2 + (k+1)^3 && (\text{by part (a)}) \\ &= (k+1)^2 \left[\frac{k^2}{4} + (k+1) \right] \\ &= (k+1)^2 \left[\frac{k^2 + 4k + 4}{4} \right] \\ &= (k+1)^2 \left[\frac{(k+2)^2}{4} \right] \\ &= \left(\frac{(k+1)(k+2)}{2} \right)^2 \\ &= (1 + 2 + \cdots + k + (k+1))^2 \quad (\text{by part (a)}) \end{aligned}$$

(d) For the base case, observe that $0^3 = 0 < \frac{1}{4} = \frac{1^4}{4} < 1 = 1^3$. For the inductive hypothesis, assume that for arbitrary fixed integer $k \geq 1$, $1^3 + 2^3 + \cdots + (k-1)^3 < \frac{k^4}{4} < 1^3 + 2^3 + \cdots + k^3$. Now, for the inductive step,

$$\begin{aligned}
 1^3 + 2^3 + \cdots + (k-1)^3 + k^3 &< \frac{k^4}{4} + k^3 \quad (\text{by inductive hypothesis}) \\
 &= \frac{k^4 + 4k^3}{4} \\
 &< \frac{k^4 + 4k^3 + 6k^2 + 4k + 1}{4} \\
 &= \frac{(k+1)^4}{4} \\
 &= \frac{k^4}{4} + \frac{4k^3 + 6k^2 + 4k + 1}{4} \\
 &< (1^3 + 2^3 + \cdots + k^3) + \frac{4k^3 + 6k^2 + 4k + 1}{4} \\
 &= (1^3 + 2^3 + \cdots + k^3) + k^3 + \frac{6k^2}{4} + k + \frac{1}{4} \\
 &< (1^3 + 2^3 + \cdots + k^3) + k^3 + 3k^2 + 3k + 1 \\
 &= (1^3 + 2^3 + \cdots + k^3) + (k+1)^3
 \end{aligned}$$

Hence, we have $1^3 + 2^3 + \cdots + k^3 < \frac{(k+1)^4}{4} < 1^3 + 2^3 + \cdots + k^3 + (k+1)^3$ as desired. $\square$

Problem 2

Solution:

We will guess that

$$1 - 4 + 9 - \cdots + (-1)^{n+1}n^2 = (-1)^{n-1}(1 + 2 + \cdots + n)$$

For the base case, observe that $(-1)^{1+1}(1)^2 = 1 = (-1)^{1-1}(1)$.

For the inductive hypothesis (IH), assume that for arbitrary fixed integer $k \geq 1$

$$1 - 4 + 9 - \cdots + (-1)^{k+1}k^2 = (-1)^{k-1}(1 + 2 + \cdots + k)$$

Now, for the inductive step,

$$\begin{aligned}
1 - \cdots + (-1)^{k+1}k^2 + (-1)^{(k+1)+1}(k+1)^2 &= (-1)^{k-1}(1+2+\cdots+k) + (-1)^{k+2}(k+1)^2 && \text{(by the IH)} \\
&= (-1)^{k-1} \left(\frac{k(k+1)}{2} \right) + (-1)^{k+2}(k+1)^2 && \text{(by 1(a))} \\
&= (-1)^{k-1}(k+1) \left[\frac{k}{2} + (-1)^3(k+1) \right] \\
&= (-1)^{k-1}(k+1) \left[\frac{k}{2} - (k+1) \right] \\
&= (-1)^{k-1}(k+1) \left[\frac{k}{2} - k - 1 \right] \\
&= (-1)^{k-1}(k+1) \left[-\frac{k}{2} - 1 \right] \\
&= (-1)^k(k+1) \left[\frac{k}{2} + 1 \right] \\
&= (-1)^k \left[\frac{k(k+1)}{2} + (k+1) \right] \\
&= (-1)^k [(1+2+\cdots+k) + (k+1)] && \text{(by 1(a))} \\
&= (-1)^{((k+1)-1)} [1+2+\cdots+k+(k+1)]
\end{aligned}$$

as desired. □

Problem 3

Solution:

We will guess that

$$1 + \left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} \right) = 2 - \frac{1}{2^n}$$

For the base case, observe that $1 + \frac{1}{2} = \frac{2+1}{2} = \frac{3}{2} = \frac{4}{2} - \frac{1}{2} = 2 - \frac{1}{2}$.

For the inductive hypothesis (IH), assume that for arbitrary fixed integer $k \geq 1$

$$1 + \left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^k} \right) = 2 - \frac{1}{2^k}$$

Now, for the inductive step,

$$\begin{aligned}
 \left[1 + \left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^k} \right) \right] + \frac{1}{2^{k+1}} &= 2 - \frac{1}{2^k} + \frac{1}{2^{k+1}} && \text{(by the IH)} \\
 &= 2 - \left(\frac{2}{2} \right) \left(\frac{1}{2^k} \right) + \left(\frac{1}{2} \right) \left(\frac{1}{2^k} \right) \\
 &= 2 - \left(\frac{1}{2^k} \right) \left(\frac{2}{2} - \frac{1}{2} \right) \\
 &= 2 - \left(\frac{1}{2^k} \right) \left(\frac{1}{2} \right) \\
 &= 2 - \frac{1}{2^{k+1}}
 \end{aligned}$$

as desired. □

Problem 4

Solution:

We will guess that $\forall n \in \mathbb{Z}^{\geq 2}$,

$$\left(1 - \frac{1}{2} \right) \left(1 - \frac{1}{3} \right) \cdots \left(1 - \frac{1}{n} \right) = \frac{1}{n}$$

For the base case, observe that $1 - \frac{1}{2} = \frac{1}{2}$.

For the inductive hypothesis (IH), assume that for arbitrary fixed integer $k \geq 2$

$$\left(1 - \frac{1}{2} \right) \left(1 - \frac{1}{3} \right) \cdots \left(1 - \frac{1}{k} \right) = \frac{1}{k}$$

Now, for the inductive step,

$$\begin{aligned}
 \left(1 - \frac{1}{2} \right) \cdots \left(1 - \frac{1}{k} \right) \left(1 - \frac{1}{k+1} \right) &= \left(\frac{1}{k} \right) \left(1 - \frac{1}{k+1} \right) && \text{(by the IH)} \\
 &= \left(\frac{1}{k} \right) \left(\frac{(k+1) - 1}{k+1} \right) \\
 &= \left(\frac{1}{k} \right) \left(\frac{k}{k+1} \right) \\
 &= \frac{1}{k+1}
 \end{aligned}$$

as desired. □

Problem 5

Solution:

We'll guess that $\forall n \in \mathbb{Z}^{\geq 2}$

$$\begin{aligned}
 \left(1 - \frac{1}{2^2}\right) \cdots \left(1 - \frac{1}{n^2}\right) &= \left[\left(1 + \frac{1}{2}\right) \left(1 - \frac{1}{2}\right)\right] \cdots \left[\left(1 + \frac{1}{n}\right) \left(1 - \frac{1}{n}\right)\right] \\
 &= \left[\left(1 + \frac{1}{2}\right) \cdots \left(1 + \frac{1}{n}\right)\right] \left[\left(1 - \frac{1}{2}\right) \cdots \left(1 - \frac{1}{n}\right)\right] \\
 &= \left[\left(1 + \frac{1}{2}\right) \cdots \left(1 + \frac{1}{n}\right)\right] \left[\frac{1}{n}\right] \quad (\text{by problem 4}) \\
 &= \left[\left(\frac{3}{2}\right) \left(\frac{4}{3}\right) \cdots \left(\frac{n}{n-1}\right) \left(\frac{n+1}{n}\right)\right] \left[\frac{1}{n}\right] \\
 &= \left[\frac{n+1}{2}\right] \left[\frac{1}{n}\right] \\
 &= \frac{n+1}{2n} \\
 &= \frac{1}{2} + \frac{1}{2n}
 \end{aligned}$$

For the base case, observe that $\left(1 - \frac{1}{4}\right) = \frac{3}{4} = \frac{1}{2} + \frac{1}{2(2)}$.

Now for the inductive hypothesis, assume that for arbitrary fixed integer $k \geq 2$,

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \cdots \left(1 - \frac{1}{k^2}\right) = \frac{1}{2} + \frac{1}{2k}$$

Finally, for the inductive step

$$\begin{aligned}
 \left(1 - \frac{1}{2^2}\right) \cdots \left(1 - \frac{1}{k^2}\right) \left(1 - \frac{1}{(k+1)^2}\right) &= \left(\frac{1}{2} + \frac{1}{2k}\right) \left(1 - \frac{1}{(k+1)^2}\right) \quad (\text{by the inductive hypothesis}) \\
 &= \left(\frac{k+1}{2k}\right) \left(\frac{(k+1)^2 - 1}{(k+1)^2}\right) \\
 &= \left(\frac{k+1}{2k}\right) \left(\frac{((k+1)+1)((k+1)-1)}{(k+1)^2}\right) \\
 &= \left(\frac{k+1}{2k}\right) \left(\frac{(k+2)k}{(k+1)^2}\right) \\
 &= \frac{k+2}{2(k+1)} \\
 &= \frac{(k+1)+1}{2(k+1)} \\
 &= \frac{k+1}{2(k+1)} + \frac{1}{2(k+1)} \\
 &= \frac{1}{2} + \frac{1}{2(k+1)}
 \end{aligned}$$

as desired.

□

Problem 6

Solution:

(a) Suppose that $A(k)$ is true, i.e.

$$1 + 2 + \cdots + k = \frac{1}{8}(2k+1)^2$$

Then, we have

$$\begin{aligned}
 1 + 2 + \cdots + k + (k + 1) &= \frac{1}{8}(2k + 1)^2 + (k + 1) \quad (\text{since } A(k) \text{ is true}) \\
 &= \frac{1}{8}(4k^2 + 4k + 1) + \frac{8(k + 1)}{8} \\
 &= \frac{4k^2 + 12k + 9}{8} \\
 &= \frac{(4k^2 + 8k + 4) + (4k + 4) + 1}{8} \\
 &= \frac{4(k^2 + 2k + 1) + 4(k + 1) + 1}{8} \\
 &= \frac{4(k + 1)^2 + 4(k + 1) + 1}{8} \\
 &= \frac{1}{8}(2(k + 1) + 1)^2
 \end{aligned}$$

which is $A(k + 1)$. Thus, $A(k) \implies A(k + 1)$.

(b) We haven't shown a base case (i.e. that $A(t)$ is true for some positive integer t) so we can't conclude that.

(c) Observe that

$$\begin{aligned}
 \frac{1}{8}(2n + 1)^2 &= \frac{1}{8} \left(2 \left(n + \frac{1}{2} \right) \right)^2 \\
 &= \frac{4}{8} \left(n + \frac{1}{2} \right)^2 \\
 &= \frac{1}{2} \left(n + \frac{1}{2} \right)^2 \\
 &= \frac{n^2 + n + \frac{1}{4}}{2} \\
 &> \frac{n^2 + n}{2} \\
 &= \frac{n(n + 1)}{2} \\
 &= 1 + 2 + \cdots + n \quad (\text{by 1(a)})
 \end{aligned}$$

We'll now use induction to show that this is true for all positive n .

For the base case, observe that $1 < \frac{9}{8} = \frac{1}{8}(2(1) + 1)^2$

For the inductive hypothesis, assume that for arbitrary fixed $k \geq 1$

$$1 + 2 + \cdots + k < \frac{1}{8}(2k + 1)^2$$

Now, for the inductive step,

$$\begin{aligned} 1 + 2 + \cdots + k + (k + 1) &< \frac{1}{8}(2k + 1)^2 + (k + 1) \\ &= \frac{1}{8}(2(k + 1) + 1)^2 \end{aligned}$$

from the proof of $A(k) \implies A(k + 1)$ in part (a).

□

Problem 7

Solution:

Observe that

$$\begin{array}{lll} (1 + x)^1 & = 1 + x & < 1 + (1)x + (1)x^2 \\ (1 + x)^2 & = 1 + x + x^2 & < 1 + (2)x + (2)x^2 \\ (1 + x)^3 & = 1 + 3x + 3x^2 + x^3 & > 1 + (3)x + (3)x^2 \\ (1 + x)^4 & = 1 + 4x + 6x^2 + 4x^3 + x^4 & > 1 + (4)x + (4)x^2 \\ \vdots & & \end{array}$$

so we hypothesize that $n_1 = 3$. We'll now prove it by induction.

We've already shown the base case (i.e $(1 + x)^3 > 1 + 3x + 3x^2$) above.

For the inductive hypothesis, assume that for arbitrary fixed integer $k \geq 3$,

$$(1 + x)^k > 1 + kx + kx^2$$

Now, for the inductive step,

$$\begin{aligned} (1 + x)^{k+1} &= (1 + x)^k(1 + x) \\ &> (1 + kx + kx^2)(1 + x) \quad (\text{by the inductive hypothesis}) \\ &= 1 + kx + kx^2 + x + kx^2 + kx^3 \\ &= 1 + (k + 1)x + (k + k)x^2 + kx^3 \\ &> 1 + (k + 1)x + (k + 1)x^2 \end{aligned}$$

as desired.

□

Problem 8

Solution:

For the base case, clearly $a_1 \leq a_1 = a_1(c^0) = a_1(c^{1-1})$

For the inductive hypothesis, for arbitrary fixed $k \geq 1$, assume that $a_k \leq a_1(c^{k-1})$.

Now, for the inductive step, observe that $a_{k+1} \leq c(a_{(k+1)-1}) = c(a_k)$ by the given lemma. The lemma applies here because $k \geq 1 \implies k+1 \geq 2$. By the inductive hypothesis, $c(a_k) \leq c(a_1(c^{k-1})) = a_1(c^k)$. Hence, we have $a_{k+1} \leq a_1(c^k)$ as desired. $\square$

Problem 9

Solution:

For the base case, we can definitely construct a line of length $1 = \sqrt{1}$ (we already have it).

For the inductive hypothesis, for arbitrary fixed $k \geq 1$ assume that we can construct a line of length $\sqrt{k}$.

For the inductive step, observe that $(\sqrt{k})^2 + 1^2 = (\sqrt{k+1})^2$. Hence, if we construct a line segment of length $\sqrt{k}$ (which we know we can do from the inductive hypothesis) and a line segment of unit length (which we already have) such that they are perpendicular to one another, the length of the hypotenuse will be precisely $\sqrt{k+1}$. $\square$

Problem 10

Solution:

For the base case, observe that $n = 0 = 0b + 0$. Hence $q = 0$ and $r = 0$ work.

For the inductive hypothesis, for arbitrary fixed integer $k \geq 0$ assume that there exist integer q, r ($0 \leq r < b$) such that $k = qb + r$.

Now, for the inductive step, observe that $n+1 = (qb+r)+1 = qb+(r+1)$ by the inductive hypothesis. Since $0 \leq r < b$, we have $0 < r+1 \leq b$. Hence, there are two cases. If $r+1 = b$, then $n+1 = qb+b = (q+1)b+0$. And if $r+1 < b$, then $n+1 = qb+(r+1)$ satisfies the theorem directly. Thus, in all cases we are able to find integers x, y ($0 \leq y < b$) such that $n+1 = xb+y$. $\square$

Problem 11

Solution:

In this problem, we'll use a variant of induction called strong induction. The difference is as follows. In "normal" / "weak" induction, we would assume $A(k)$ for arbitrary fixed integer $k \geq n_0$ (n_0 is the base case) and try to show $A(k+1)$. However, in strong induction, we'll assume $A(n_0+1), A(n_0+2), \dots, A(k)$ for arbitrary fixed integer $k \geq n_0$ and try to show $A(k+1)$. That is, the inductive hypothesis is stronger than before.

Let's start.

For the base case, we will show that $n = 2$ is prime. Since $2 = 1(2)$, both 1, 2 divide n . Also, no integer $d > 2$ can divide 2 – for any positive integer c we have $cd > 2c > 2$, for any negative integer c we have $cd < 0 < 2$, and for $c = 0$, we have $cd = 0d = 0 < 2$. Thus, 1 and $n = 2$ are the only divisors of $n = 2$, so 2 is prime.

For the inductive hypothesis, for arbitrary fixed integer k , assume that every one of $2, 3, \dots, k$ is either prime or the product of primes.

Now, for the inductive step, we have two cases. If $k+1$ is prime, then we're done.

Otherwise, there exist integers c, d (at least one of which is not 1 or $k+1$) such that $cd = k+1$. Without loss of generality, we can assume c, d are both positive. If both are negative then $(-c), (-d)$ are both positive and $(-c)(-d) = cd = k+1$. And if exactly one is negative then $cd < 0 < k+1$ (a contradiction).

Now, we will show that both $1 < c < k+1$ and $1 < d < k+1$ must be true.

As stated above, at least one of c, d is not 1 or $k+1$ (but both are positive). Hence, without loss of generality, we can say that $1 < c < k+1$. There are 3 cases for d – $d = 1$, $1 < d < k+1$, or $d \geq k+1$. Observe that $d = 1$ doesn't work because otherwise $cd < (k+1)(1) = k+1$. And if $d \geq k+1$ then $cd > 1(k+1) = k+1$. Thus, it must also be true that $1 < d < k+1$.

But now, by the inductive hypothesis, we can say that c, d are each either prime or a product of primes. Thus, we can conclude that $cd = k+1$ is a product of primes (namely those that make up c, d).

□

Problem 12

Solution:

Let $A(k)$ be the proposition that given k girls with at least one of them blue-eyed, all k of them are blue-eyed. The proof holds that $A(k) \implies A(k+1)$ by illustrating $A(3) \implies A(4)$ (and claiming all other $k \geq 1$ follow similar logic). However, the proof breaks down because it's *not* true that all other k are similar.

Consider the following case: we have 2 girls (say G_1, G_2) and only 1 of them (e.g. G_1) is blue-eyed. Now, we cannot invoke the inductive hypothesis on G_2 alone since it's not applicable, i.e. we have a set of 1 girls, but there is no blue-eyed girl. Thus, $A(1) \not\Rightarrow A(2)$, breaking the induction chain.

□